Seminar 4 - Matrix Decomposition and Vector Calculus

ECE 447: Data Analysis and Machine Learning for Engineers

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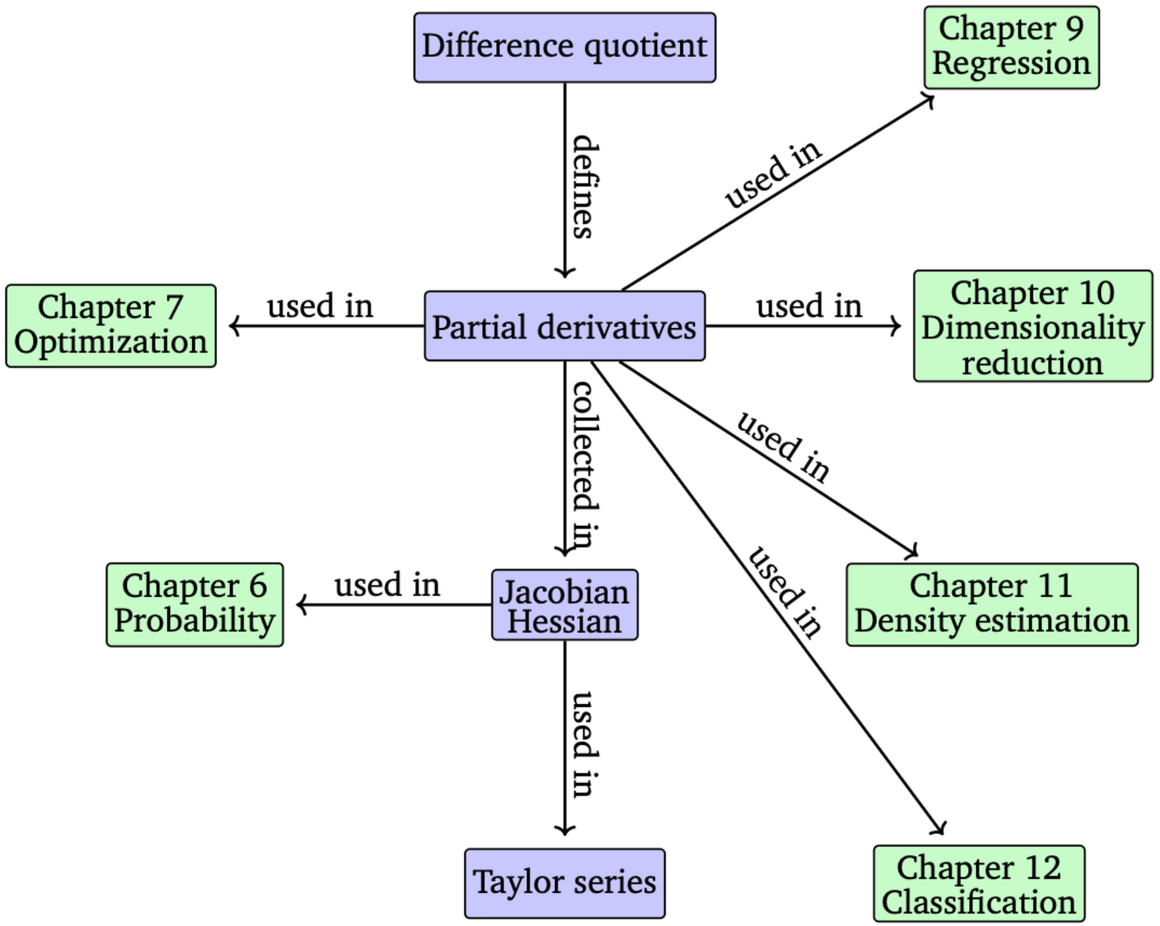
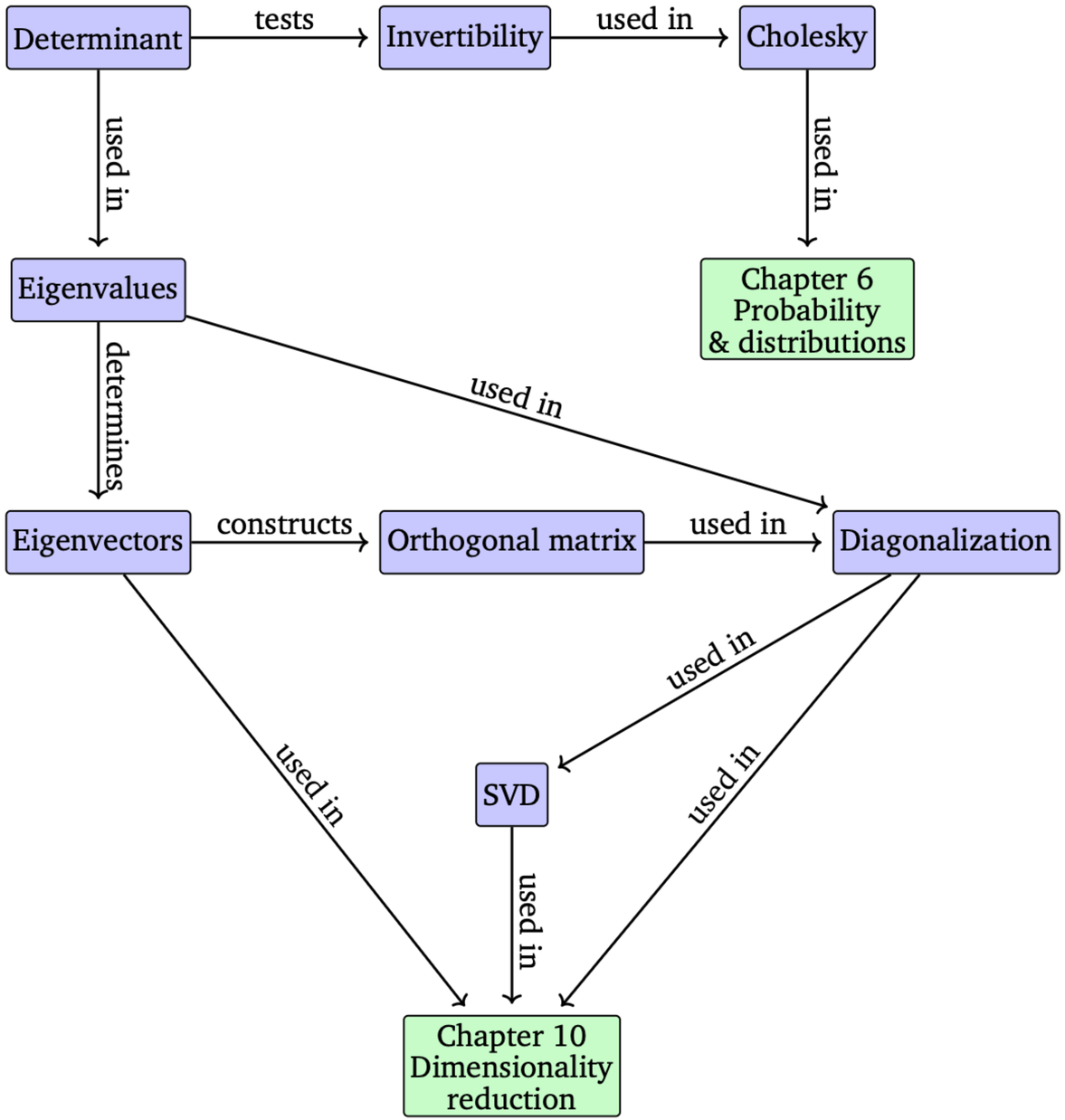
This Seminar Covers





Eigenvalues and Eigenvectors

Characterize a matrix and its associated linear mapping.

Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be a square matrix. Then $\lambda \in \mathbb{R}$ is an *eigenvalue* of $\boldsymbol{A}$ and $\boldsymbol{x} \in \mathbb{R}^n \backslash \{\boldsymbol{0}\}$ is the corresponding *eigenvector* of $\boldsymbol{A}$ if

$$\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}$$

This is the eigenvalue equation.

Collinear and Codirection

Codirection

Two vector that point in the same direction.

Collinear

Two vector that point in the same or opposite direction.

Remark. If $\boldsymbol{x}$ is an eigenvector of $\boldsymbol{A}$ associated with eigenvalue λ, then for any $c \in \mathbb{R}\backslash\{0\}$ it holds that $c\boldsymbol{x}$ is an eigenvector of $\boldsymbol{A}$ with the same eigenvalue, i.e.,

$$\boldsymbol{A}(c\boldsymbol{x}) = c\boldsymbol{A}\boldsymbol{x} = c\lambda\boldsymbol{x} = \lambda(c\boldsymbol{x})$$

Eigenspace and EigenSpectrum

Eigenspace

The subspace in $\mathbb{R}^n$ spanned by the set of all eigenvectors of $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ associated with an eigenvalue λ.

Eigenspectrum

The set of all eigenvalues of $\boldsymbol{A}$, a.k.a. spectrum of $\boldsymbol{A}$.

Remark. The eigenspace E_λ is the solution space of the system of linear equations

$$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}$$

Geometric Intuition on Eigenvalues

An *eigenvectors* corresponding to a nonzero eigenvalue points in a direction that is stretched by the linear mapping.

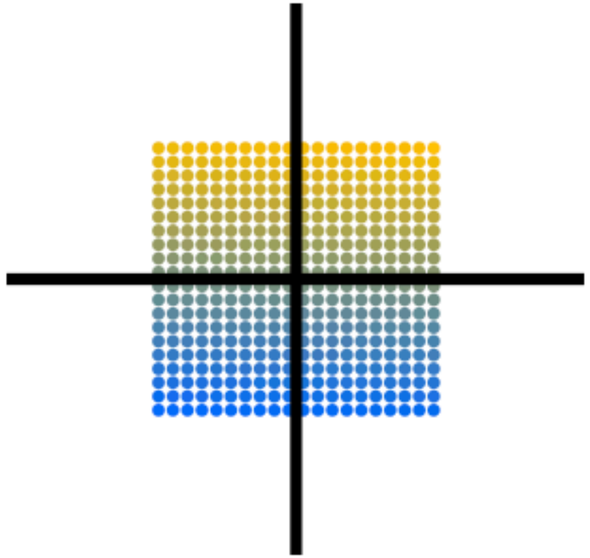
The eigenvalue is a factor by which it is stretched. A negative eigenvalue means the direction of the stretching is flipped.

So eigenvectors and eigenvalues are characterizing the behaviour of the matrix $\boldsymbol{A}$ as a linear map.

Ex1. Canonical Basis Vector

$$\boldsymbol{A} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{bmatrix}$$

Input



$$\lambda_1 = 2.0$$
$$\lambda_2 = 0.5$$
$$\det(\boldsymbol{A}) = 1.0$$

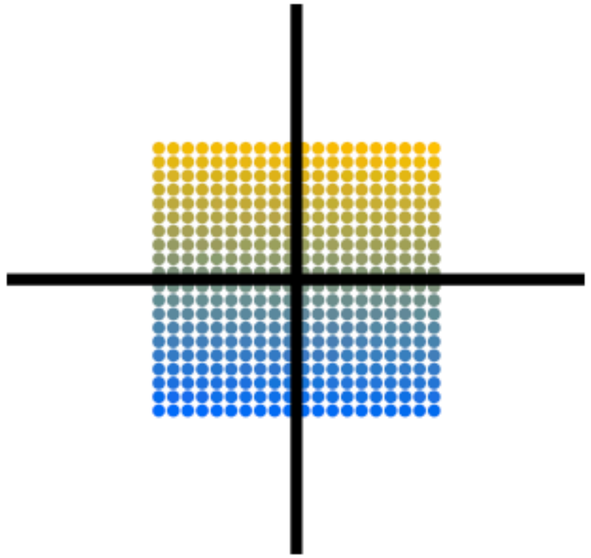
Output

Extend the vertical axis, and compress the horizontal.

Ex2. Shears

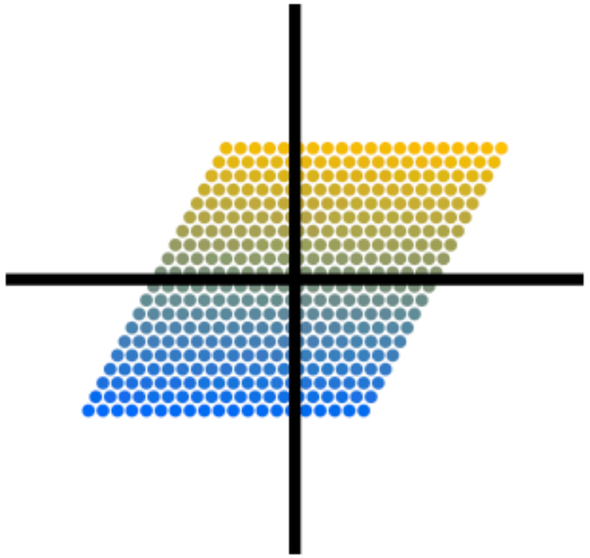
$$\boldsymbol{A} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

Input



$$\lambda_1 = 1.0$$
$$\lambda_2 = 1.0$$
$$\det(\boldsymbol{A}) = 1.0$$

Output

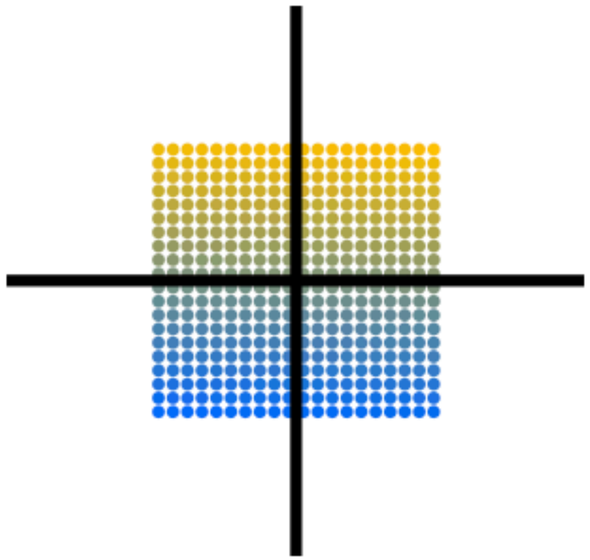


Mapping acts only along one direction (the horizontal axis).

Ex3. Rotation

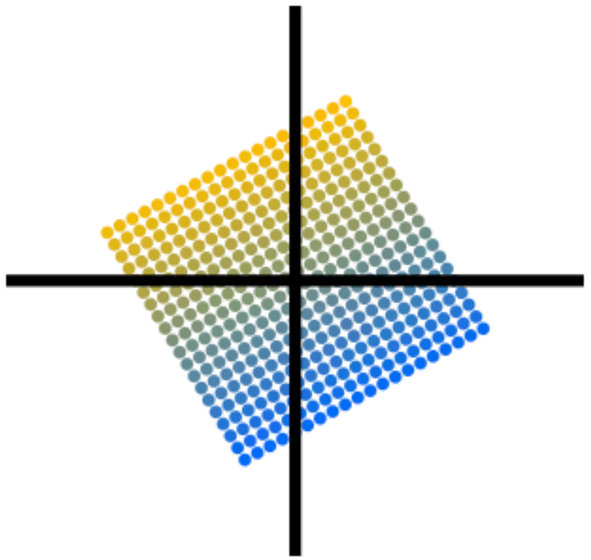
$$\boldsymbol{A} = \begin{bmatrix} \cos(\frac{\pi}{6}) & -\sin(\frac{\pi}{6}) \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) \end{bmatrix}$$

Input



$$\lambda_1 = (0.87\text{-}0.5\text{j})$$
$$\lambda_2 = (0.87\text{+}0.5\text{j})$$
$$\det(\boldsymbol{A}) = 1.0$$

Output

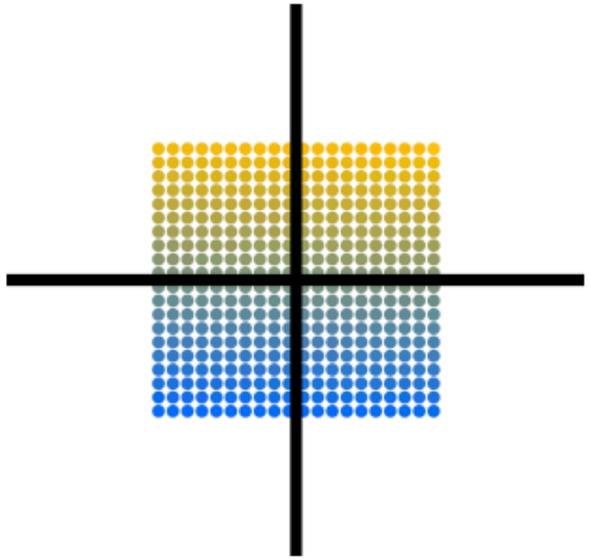


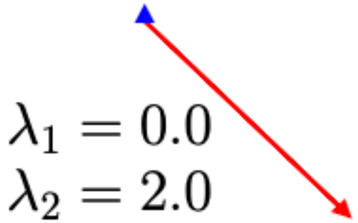
Rotating the space.

Ex4. Collapse

$$\boldsymbol{A} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

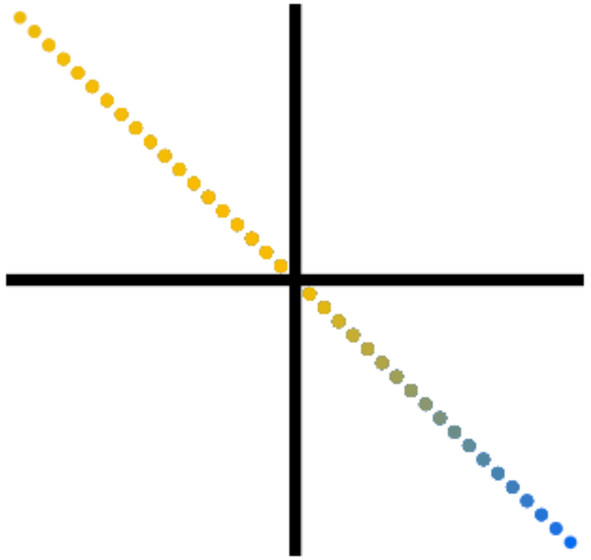
Input





$\lambda_1 = 0.0$
$\lambda_2 = 2.0$
$\det(\boldsymbol{A}) = 0.0$

Output

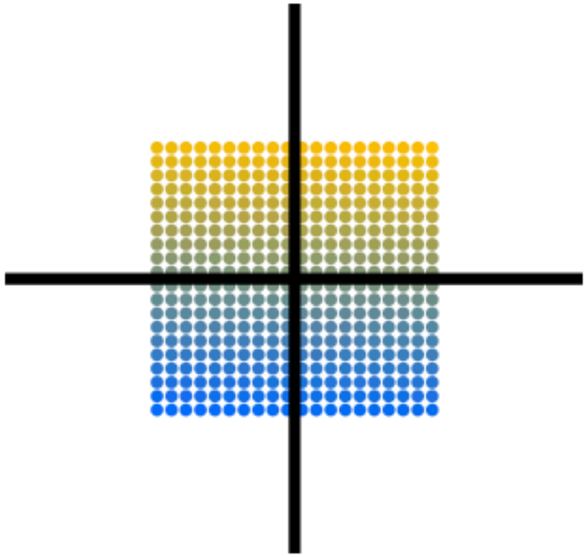


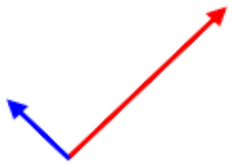
Collapsing a two-dimensional domain onto one dimension.

Ex5. Shear and Stretch

$$\boldsymbol{A} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}$$

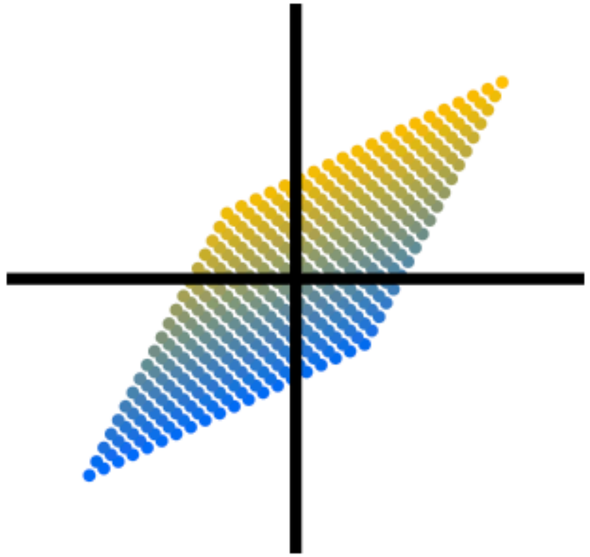
Input





$$\lambda_1 = 0.5$$
$$\lambda_2 = 1.5$$
$$\det(\boldsymbol{A}) = 0.75$$

Output



Spectral Theorem

The eigenvectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ with n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ are linearly independent.

*They form a **basis** of $\mathbb{R}^n$.*

Spectral Theorem

*If $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is symmetric, there exists an **orthonormal basis** of the corresponding vector space V consisting of **eigenvectors** of $\boldsymbol{A}$, and each eigenvalue is real.*

We Can Decompose a Symmetric Matrix

Spectral theorem implies that we can perform eigendecomposition of a *symmetric* matrix $\boldsymbol{A}$ (with real eigenvalues) as follows

$$\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^\top$$

where

$\boldsymbol{\Lambda}$: diagonal matrix containing eigenvalues $\boldsymbol{\Lambda}_{ii} = \lambda_i$.

$\boldsymbol{V}$: contain eigenvectors as its columns (this is an orthogonal basis).

This is an important and useful property in this course. Why?

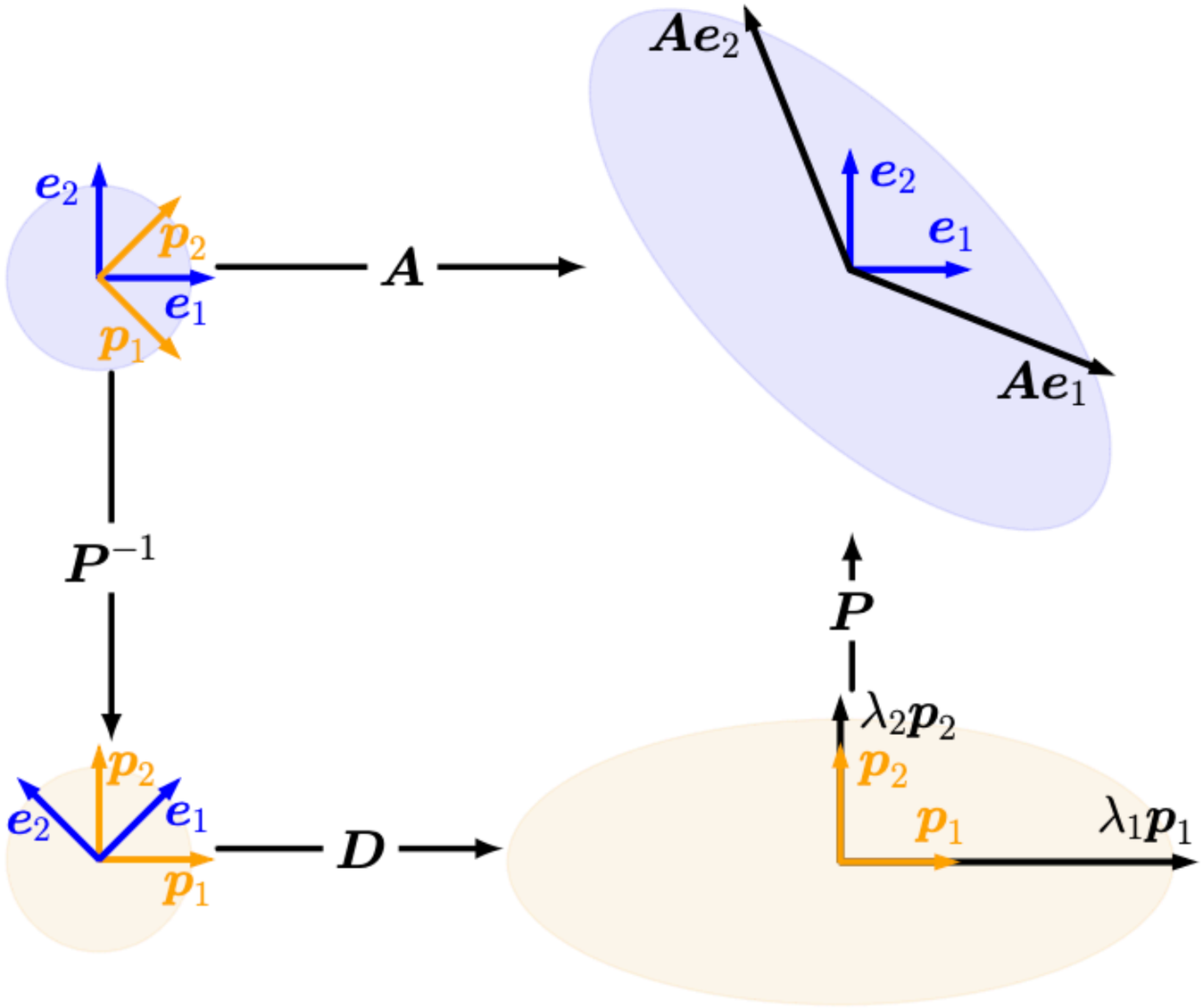
Eigendecomposition

A square matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ can be factored into

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$$

where $\boldsymbol{P} \in \mathbb{R}^{n \times n}$ and $\boldsymbol{D}$ is a diagonal matrix whose diagonal entries are the eigenvalues of $\boldsymbol{A}$, if and only if the eigenvectors of $\boldsymbol{A}$ form a basis of $\mathbb{R}^n$.

Eigendecomposition as Sequential Transformations Visualized



Diagonalizable

A matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is *diagonalizable* if it is similar to a diagonal matrix, i.e., if there exists an invertible matrix $\boldsymbol{P} \in \mathbb{R}^{n \times n}$ such that

$$\boldsymbol{D} = \boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P}$$

What is this doing? We are expressing the same linear mapping $\boldsymbol{A}$ but in another basis.

What is that basis? The eigenvectors of $\boldsymbol{A}$.

Note: $\boldsymbol{P}\boldsymbol{D} = \boldsymbol{A}\boldsymbol{P}$

Decompositions are Useful in Machine Learning

Cholesky Decomposition

A symmetric, positive definite matrix $\boldsymbol{A}$ can be factorized into a product

$$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^\top$$

$\boldsymbol{L}$ is the Cholesky factor of $\boldsymbol{A}$, and $\boldsymbol{L}$ is unique.

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n1} & \cdots & l_{nn} \end{bmatrix} \begin{bmatrix} l_{11} & \cdots & l_{n1} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & l_{nn} \end{bmatrix}$$

where $\boldsymbol{L}$: is a lower triangular matrix with positive diagonal elements.

Cholesky Decomposition in Machine Learning

Useful for sampling from multivariate normal $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

If $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, then

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{\mu}$$

So choose an $\boldsymbol{A}$ using Cholesky decomposition on $\boldsymbol{\Sigma}$, and do linear transformation on normal samples.

where $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{\Sigma}$.

$\boldsymbol{y}$ is Gaussian distributed with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.

So far all we have looked into
were **square** matrices,
what if our matrix is **rectangular**?

Singular Value Decomposition (SVD)

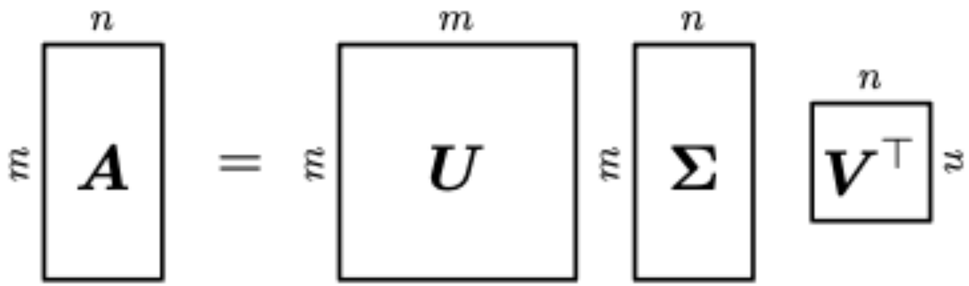
SVD Theorem

Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ *be a rectangular matrix of rank* $r \in [0, \min(m, n)]$. *The SVD of* $\boldsymbol{A}$ *is a decomposition of the form*

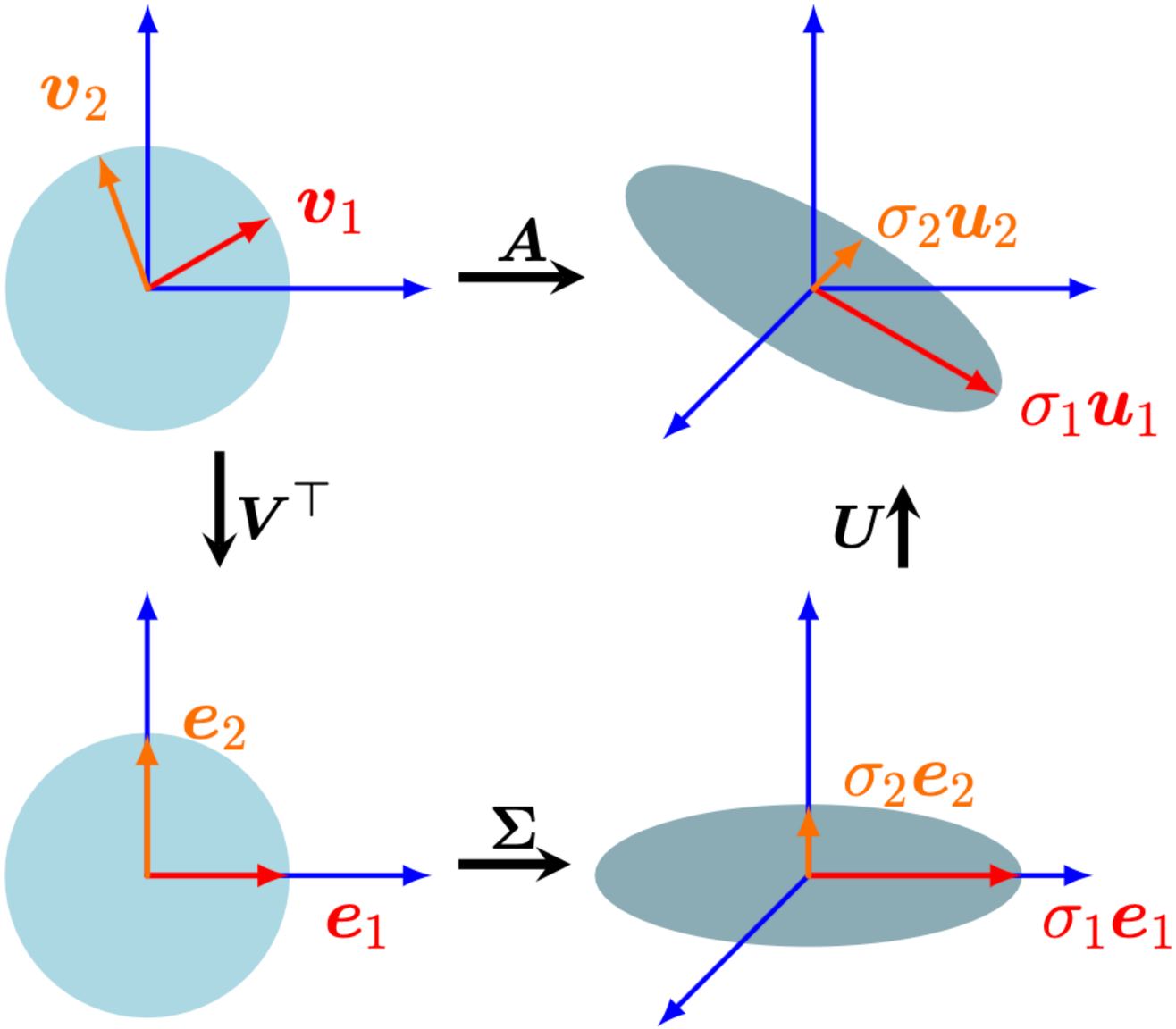
$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$$

where $\boldsymbol{U}$ and $\boldsymbol{V}$ are orthogonal matrices, and $\boldsymbol{\Sigma}$ is a diagonal matrix with $\Sigma_{ii} = \sigma_i \geq 0$.

Singular values



Singular Value Decomposition Visualized



Relation between Eigenvalues and Singular Values

Consider

$$\boldsymbol{A}^\top \boldsymbol{A}$$

$$(\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top)^\top(\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top)$$

$$\boldsymbol{V}\boldsymbol{\Sigma}^\top\boldsymbol{U}^\top\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$$

$$\boldsymbol{V}\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma}\boldsymbol{V}^\top$$

$$\boldsymbol{V}\begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_n^2 \end{bmatrix}\boldsymbol{V}^\top$$

Hence, $\sigma_i^2 = \lambda_i$.

Eigenvalue Decompositions vs. Singular Value Decompositions

Consider the eigendecomposition $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$ and the SVD $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$.

- The SVD always exists for any matrix $\mathbb{R}^{m\times n}$. The eigendecomposition is only defined for square matrices $\mathbb{R}^{n\times n}$ and only exists if we can find a basis of eigenvectors of $\mathbb{R}^n$.

- The vectors in the eigendecomposition matrix $\boldsymbol{P}$ are not necessarily orthogonal, i.e., the change of basis is not a simple rotation and scaling. On the other hand, the vectors in the matrices $\boldsymbol{U}$ and $\boldsymbol{V}$ in the SVD are orthonormal, so they do represent rotations.

- Both the eigendecomposition and the SVD are compositions of three linear mappings:
  1. Change of basis in the domain.
  2. Independent scaling of each new basis vector and mapping from domain to codomain.
  3. Change of basis in the codomain.

Rules of Partial Differentiation

Product Rule

$$\frac{\partial}{\partial \boldsymbol{x}}(f(\boldsymbol{x})g(\boldsymbol{x})) = \frac{\partial f}{\partial \boldsymbol{x}}g(\boldsymbol{x}) + f(\boldsymbol{x})\frac{\partial g}{\partial \boldsymbol{x}}$$

Sum Rule

$$\frac{\partial}{\partial \boldsymbol{x}}(f(\boldsymbol{x}) + g(\boldsymbol{x})) = \frac{\partial f}{\partial \boldsymbol{x}} + \frac{\partial g}{\partial \boldsymbol{x}}$$

Chain Rule

$$\frac{\partial}{\partial \boldsymbol{x}}(g \circ f)(\boldsymbol{x}) = \frac{\partial}{\partial \boldsymbol{x}}(g(f(\boldsymbol{x}))) = \frac{\partial g}{\partial f}\frac{\partial f}{\partial \boldsymbol{x}}$$

Gradient of a Function with Respect to A Variable

Consider a function $f : \mathbb{R}^2 \to \mathbb{R}$ of two variables x_1, x_2, which themselves are functions of t.

$$\frac{df}{dt} = [\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2}] \begin{bmatrix} \frac{\partial x_1(t)}{\partial t} \\ \frac{\partial x_2(t)}{\partial t} \end{bmatrix} = \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial t}$$

where d denotes the gradient and ∂ partial derivatives.

Gradients of Vector-Values Functions

For a function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, and a vector $\boldsymbol{x} = [x_1, \ldots, x_n]^\top \in \mathbb{R}^n$, we have

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} f_1(\boldsymbol{x}) \\ \vdots \\ f_m(\boldsymbol{x}) \end{bmatrix} \in \mathbb{R}^m$$

where $f_i : \mathbb{R}^n \to \mathbb{R}$. Then,

$$\frac{d\boldsymbol{f}(\boldsymbol{x})}{d\boldsymbol{x}} = [\frac{\partial \boldsymbol{f}(\boldsymbol{x})}{\partial x_1} \quad \frac{\partial \boldsymbol{f}(\boldsymbol{x})}{\partial x_2} \quad \cdots \quad \frac{\partial f(\boldsymbol{x}}{\partial x_n}] \in \mathbb{R}^{m \times n}$$

Jacobian

The collection of all first-order partial derivatives of a vector-valued function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ is called the *Jacobian*.

$$\boldsymbol{J} = \nabla_{\boldsymbol{x}} \boldsymbol{f} = \frac{d\boldsymbol{f}(\boldsymbol{x})}{d\boldsymbol{x}} = [\frac{\partial \boldsymbol{f}(\boldsymbol{x})}{\partial x_1} \quad \frac{\partial \boldsymbol{f}(\boldsymbol{x})}{\partial x_2} \quad \cdots \quad \frac{\partial f(\boldsymbol{x}}{\partial x_n}] \in \mathbb{R}^{m \times n}$$

$$\boldsymbol{J} = \begin{bmatrix} \frac{\partial f_1(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\boldsymbol{x})}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_m(\boldsymbol{x})}{\partial x_n} \end{bmatrix}$$

Ex1. Gradient of a Vector-Valued Function

Let $\boldsymbol{f} = \boldsymbol{A}\boldsymbol{x}$, where $\quad \boldsymbol{f}(\boldsymbol{x}) \in \mathbb{R}^m, \quad \boldsymbol{A} \in \mathbb{R}^{m\times n}, \quad \boldsymbol{x} \in \mathbb{R}^n.$

$$\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m, \qquad \frac{d\boldsymbol{f}(\boldsymbol{x})}{d\boldsymbol{x}} \in \mathbb{R}^{m\times n}$$

$$f_i(\boldsymbol{x}) = \sum_{j=1}^{n} A_{ij}x_j \quad \Rightarrow \quad \frac{\partial f_i}{\partial x_j} = A_{ij} \qquad \frac{d\boldsymbol{f}}{d\boldsymbol{x}} = \begin{bmatrix} \frac{\partial f_1(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\boldsymbol{x})}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_m(\boldsymbol{x})}{\partial x_n} \end{bmatrix} = \boldsymbol{A}$$

Gradient of a Least-Squares Loss in a Linear Model

Let $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{x} \in \mathbb{R}^p$, and $\boldsymbol{A} \in \mathbb{R}^{n \times p}$. We define the loss functions as

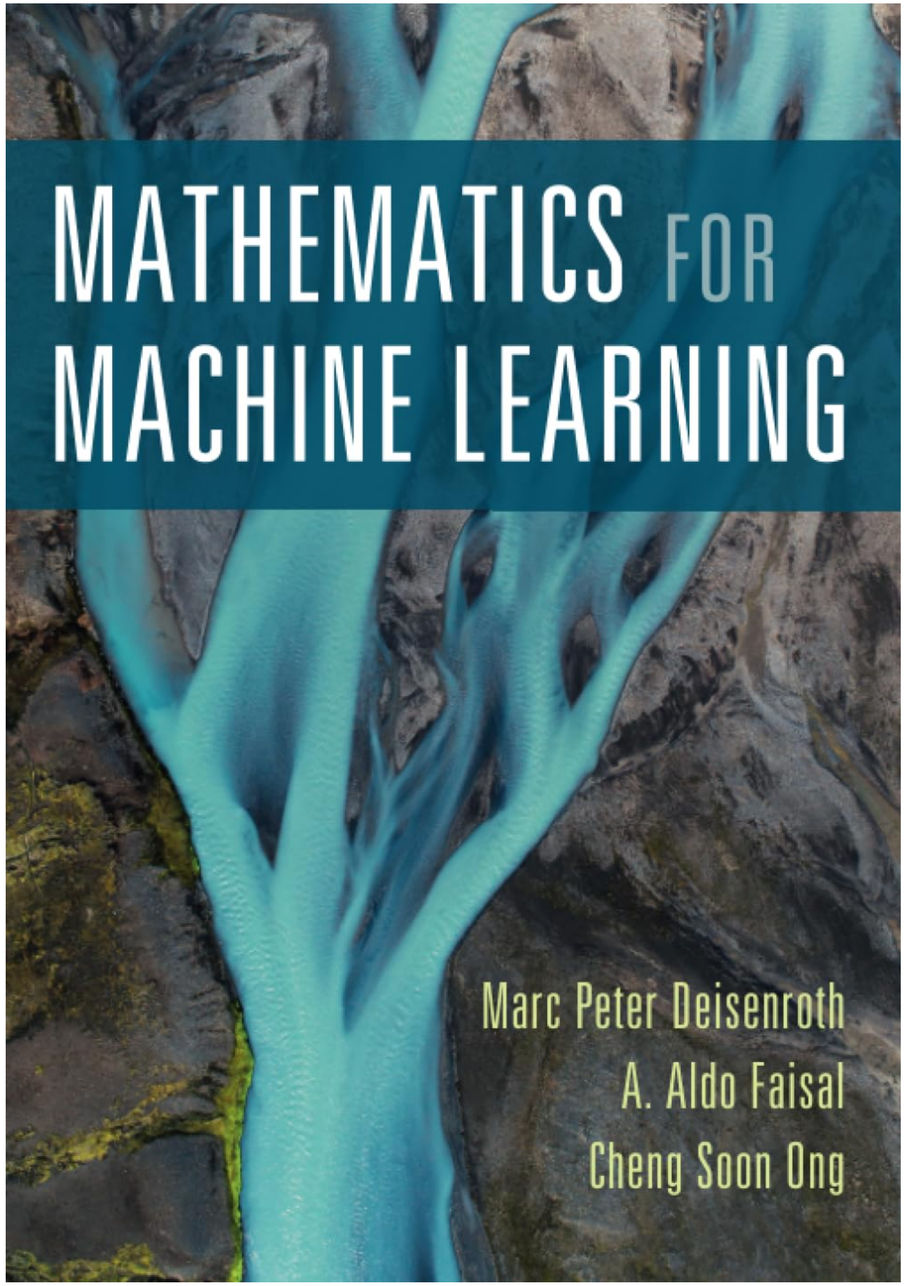
$$\mathsf{L}(\boldsymbol{e}) := \|\boldsymbol{e}\|_2^2, \qquad \boldsymbol{e}(\boldsymbol{x}) := \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$$

Compute the gradient $\dfrac{\partial L}{\partial \boldsymbol{x}}$.

$$\frac{\partial L}{\partial \boldsymbol{x}} \in \mathbb{R}^{1 \times p} \qquad \frac{\partial L}{\partial \boldsymbol{x}} = \frac{\partial L}{\partial \boldsymbol{e}} \frac{\partial \boldsymbol{e}}{\partial \boldsymbol{x}}$$

$$\left.\begin{aligned} \frac{\partial \boldsymbol{e}}{\partial \boldsymbol{x}} &= -\boldsymbol{A} \in \mathbb{R}^{n \times p} \\ \frac{\partial L}{\partial \boldsymbol{e}} &= 2\boldsymbol{e}^\top \in \mathbb{R}^{1 \times n} \end{aligned}\right. \Rightarrow \quad \frac{\partial L}{\partial \boldsymbol{x}} = -2\boldsymbol{e}^\top \boldsymbol{A} = -2(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x})^\top \boldsymbol{A} \in \mathbb{R}^{1 \times p}$$

Wanna be A Machine Learning Expert?

Learn *the math behind machine learning!*

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